

Statistical Inference in Redistricting: Challenges and Framework

S.0 — Statistical Inference Track

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Abstract

Redistricting analysis has adopted the vocabulary of statistical inference without the infrastructure: ensembles produce percentiles, not p-values; analysts report “the enacted plan is more extreme than 99% of random plans” without specifying a null distribution or correcting for multiple comparisons. This paper identifies four challenges unique to redistricting inference and proposes a three-component framework that makes redistricting claims litigation-grade.

The four challenges are: (C1) the test statistic is a function of the entire plan (a complex combinatorial object, not a real number), so standard asymptotic theory does not apply; (C2) the null distribution requires plan simulation, and simulation quality (chain convergence, ensemble size) determines inference validity; (C3) multiple metrics are evaluated on each plan (compactness, efficiency gap, mean-median, partisan bias, seat-votes curve, and others), creating a multiple testing problem that standard practice ignores; and (C4) ensembles are typically too small for reliable tail estimation—the 5th percentile of a 1,000-plan ensemble has standard error of approximately 1.5 percentage points.

The three-component framework addresses each challenge: (F1) a calibrated null distribution using the G.4 ESS-corrected ensemble that accounts for chain autocorrelation; (F2) FDR-corrected test statistics using the Benjamini-Hochberg procedure for

the full test battery; and (F3) a power analysis (S.3) establishing minimum ensemble sizes for 80% power to detect a 10-percentage-point partisan advantage.

The S-track (S.1–S.4) implements this framework with worked examples on North Carolina, Wisconsin, Georgia, Pennsylvania, and Texas.

Keywords

Statistical inference, redistricting ensemble, gerrymandering detection, null distribution, multiple testing, FDR control, power analysis, effective sample size, calibrated p-value

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1. Introduction

The redistricting ensemble literature has produced a rich empirical tradition: Markov chain Monte Carlo methods generate thousands of alternative plans, and an enacted plan’s position in this distribution characterises its extremity. If the enacted plan’s Democratic seat count falls in the 99th percentile of the ensemble, it is described as extreme. If it falls at the median, it is described as neutral.

This framework is descriptive, not inferential. It does not provide p-values, confidence intervals, or power guarantees. It does not correct for multiple comparisons when dozens of metrics are evaluated. It does not tell a court whether the observed extremity is statistically significant at any specified level.

These are not abstract methodological concerns. Courts operating under *Rucho v. Common Cause* [ruc, 2019] at the federal level have rejected partisan gerrymandering claims partly because the plaintiffs could not identify a judicially manageable standard. State courts, which remain the principal venue for partisan gerrymandering litigation, are increasingly demanding formal statistical evidence rather than intuitive percentile claims.

The expert who says “the enacted plan is more extreme than 99% of random plans” will face the question: how many plans were in the ensemble? Were the random plans drawn from a converged Markov chain? What is the margin of error on that 99% estimate? Did you correct for the fact that you tested eight different metrics?

The S-track answers these questions with a unified statistical framework.

1.1. The State of Current Practice

A survey of redistricting expert reports filed in state courts between 2016 and 2024 [Chen and Rodden, 2013, Herschlag et al., 2020] reveals four systematic gaps:

1. **No convergence certification:** Expert reports rarely compute Gelman-Rubin $\hat{R}$ or effective sample size (ESS) for their ensembles. G.4 [Della-Libera, 2026a] shows that a typical 10,000-step NC ensemble has $\text{ESS} \approx 695$ for Polsby-Popper compactness—the nominal sample size overstates the effective sample by $14\times$.
2. **No calibrated p-values:** Percentile claims are point estimates with sampling uncertainty. An ensemble with $\text{ESS} \approx 695$ and a plan at the 0.3rd percentile has a 95%

confidence interval on the true percentile of approximately [0.0%, 1.2%]. The confidence interval matters: a plan at the 1.2% boundary is less extreme than one at 0.0%.

3. **No multiple testing correction:** A typical redistricting analysis evaluates 7–12 metrics (compactness, efficiency gap, mean-median, partisan bias, etc.). At $\alpha = 0.05$, a 7-metric analysis expects 0.35 false positives per state—which is small. Across 50 states and 3 cycles, the expected number of false positives grows to 52 (Section 2.3).
4. **No power analysis:** Expert reports do not typically specify the probability that the ensemble would detect a genuine gerrymander of a given severity if one were present. S.3 [Della-Libera, 2026c] shows that a 10,000-step NC ensemble has 80% power to detect a 10pp partisan advantage but only 45% power for California.

1.2. Road Map

Section 2 analyses the four challenges in detail. Section 3 proposes the three-component framework. Section 4 describes the S-track contributions. Section 5 concludes.

2. Four Challenges Unique to Redistricting Inference

2.1. Challenge C1: The Test Statistic Is a Plan

In standard statistical testing, the test statistic is a scalar function of the data (e.g., a sample mean, a regression coefficient). Asymptotic distribution theory—central limit theorems, bootstrap consistency results—applies to such scalars.

In redistricting, the object of inference is a plan: an assignment of $n \approx 84,000$ census tracts to k districts. Summary statistics (compactness, seat counts, efficiency gap) are derived from this assignment, but they do not capture the full geometry of the plan. Two plans with identical compactness scores can be geometrically completely different.

This creates a dimensionality problem: the “parameter space” of redistricting plans is discrete and of dimension $O(n^k)$ —astronomically large. Standard asymptotic theory does not apply. Inference must proceed through simulation (the ensemble), and the quality of inference depends entirely on the quality of the simulation.

2.2. Challenge C2: The Null Distribution Requires Simulation

The null distribution in redistricting testing is the distribution of summary statistics (compactness, seat counts) across the space of all valid plans—plans that are contiguous, population-balanced, and otherwise legally compliant.

This distribution cannot be computed analytically. The space of valid plans is too large to enumerate. Instead, it is approximated by the ensemble: a set of plans drawn from a Markov chain whose stationary distribution is uniform over valid plans.

The quality of this approximation depends on:

1. **Convergence:** Has the chain mixed? Does the empirical distribution of the ensemble approximate the true stationary distribution? G.4 [Della-Libera, 2026a] shows that convergence requires 5,000–50,000 steps depending on the state.
2. **ESS:** What is the effective sample size, accounting for autocorrelation? A 10,000-step chain with lag-1 autocorrelation $\rho_1 = 0.87$ (as in NC from G.4) has $ESS = \frac{10,000}{1+2 \cdot 0.87/(1-0.87)} \approx 695$ independent samples—not 10,000.

Using the nominal sample size (10,000) instead of the ESS (695) overstates the precision of percentile estimates by a factor of $\sqrt{10,000/695} \approx 3.8$.

2.3. Challenge C3: Multiple Metrics Create a Testing Problem

A standard redistricting analysis evaluates multiple metrics simultaneously: Polsby-Popper compactness (or Reock), efficiency gap, mean-median, partisan bias, the seats-votes curve slope, Democratic seat count, and potentially others.

If each metric is tested at level $\alpha = 0.05$, the family-wise error rate (probability of at least one false positive) is:

$$FWER = 1 - (1 - \alpha)^m \approx 1 - e^{-m\alpha},$$

where m is the number of tests. For $m = 7$ metrics: $FWER = 1 - (0.95)^7 \approx 0.30$. Across 50 states and 3 cycles (1,050 tests): expected false positives = $1050 \times 0.05 = 52.5$ at the uncorrected $\alpha = 0.05$.

The standard practice of reporting “the enacted plan is a significant outlier on [metric X] at the 5% level” without adjustment for multiple comparisons produces a substantial false-positive risk.

2.4. Challenge C4: Ensembles Are Too Small for Tail Estimation

The most extreme gerrymandering claims typically involve enacted plans at the 1st or 99th percentile of the ensemble. But tail percentiles have large sampling uncertainty.

Let $\hat{p}$ be the empirical percentile of the enacted plan in an ensemble of size n^* (ESS-corrected). The standard error of $\hat{p}$ near the tails is approximately:

$$\text{SE}(\hat{p}) \approx \sqrt{\frac{p(1-p)}{n^*}}.$$

For $p = 0.01$ (1st percentile) and $n^* = 695$ (NC from G.4):

$$\text{SE}(0.01) \approx \sqrt{\frac{0.01 \times 0.99}{695}} \approx 0.0038 = 0.38 \text{ percentage points}.$$

The 95% confidence interval on the true percentile is approximately $[0.01 - 1.96 \times 0.38\%, 0.01 + 1.96 \times 0.38\%] = [0.26\%, 1.74\%]$.

This interval is not wide for a 1% claimed percentile. But for a plan at the claimed 0.1st percentile with the same ESS, the confidence interval includes 0%—meaning the plan might not actually be an outlier in the true stationary distribution.

S.3 [Della-Libera, 2026c] derives the minimum ensemble size for reliable tail estimation at each significance level.

3. The Three-Component Framework

The S-track proposes a three-component framework for redistricting statistical inference that addresses each of the four challenges.

3.1. Component F1: Calibrated Null Distribution

3.1.1. ESS-Corrected Ensemble

Rather than using the nominal sample size n as the basis for percentile confidence intervals, Component F1 requires computing the ESS-corrected sample size $n^* = \text{ESS}$ from G.4's diagnostics.

The ESS for a chain with lag-1 autocorrelation ρ_1 and n steps is:

$$n^* = \frac{n}{1 + 2 \sum_{k=1}^{\infty} \rho_k} \approx \frac{n}{1 + 2\rho_1/(1 - \rho_1)}$$

using the geometric decay approximation $\rho_k \approx \rho_1^k$.

3.1.2. Calibrated p -Value

Given the ESS-corrected sample size n^* and the enacted plan's empirical percentile $\hat{p}$, the calibrated p-value is:

$$p_{\text{cal}} = 2 \min(\hat{p}, 1 - \hat{p}) \quad \text{with 95\% CI: } \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n^*}}.$$

This replaces the percentile claim (“plan is at the 0.3rd percentile”) with a calibrated inferential claim (“plan is significantly below the expected distribution at $p_{\text{cal}} < 0.006$ with 95% CI [0.2%, 1.7%]”).

3.2. Component F2: FDR-Corrected Test Statistics

3.2.1. Benjamini-Hochberg Procedure

For a battery of m tests with raw p-values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$, the Benjamini-Hochberg procedure [Benjamini and Hochberg, 1995] controls the false discovery rate (FDR) at level q by rejecting all hypotheses $H_{(i)}$ for which:

$$p_{(i)} \leq \frac{i}{m} q.$$

This is less conservative than the Bonferroni correction (which controls the family-wise error rate) and more appropriate when many hypotheses are expected to be true simultaneously (as in a multi-state, multi-metric redistricting analysis where many states do have partisan maps).

3.2.2. Application to the Full Test Battery

For the L-track's 7 metrics $\times$ 50 states $\times$ 3 cycles = 1,050 tests at $q = 0.10$: S.4 [Della-Libera, 2026d] shows that BH correction reduces significant findings from 147 (uncorrected

at $\alpha = 0.05$) to 89, while preserving all clear gerrymanders (NC, WI, TX enacted maps) and correctly demoting borderline cases.

3.3. Component F3: Power Analysis

3.3.1. Definition

The power of a redistricting test is the probability of correctly rejecting the null hypothesis (“this plan is a random draw from valid plans”) when the alternative is true (“this plan was gerrymandered to produce a δ -percentage-point partisan advantage”).

3.3.2. Power-ESS Relationship

S.3 [Della-Libera, 2026c] derives the relationship between ensemble size, ESS, and power for detecting a 10pp partisan advantage at $\alpha = 0.05$:

$$\text{Power}(\text{ESS}, \delta) = \Phi\left(z_{\delta/2} \cdot \sqrt{\text{ESS}} \cdot \sigma_0^{-1} - z_{\alpha/2}\right)$$

where σ_0 is the null distribution standard deviation and Φ is the standard normal CDF.

This formula yields the minimum ESS for 80% power, which S.3 translates into state-specific minimum ensemble sizes (Section 4).

4. S-Track Contributions

The S-track comprises four papers (S.1–S.4) that implement the three-component framework with worked examples.

4.1. S.1: Hypothesis Testing

S.1 develops a permutation test framework for partisan gerrymandering that converts G.1’s ensemble percentiles into calibrated p-values. The key step is accounting for ensemble autocorrelation (C2) and tail-estimation uncertainty (C4) simultaneously.

For North Carolina 2022 enacted plan:

- Nominal percentile (G.1): 0.4th percentile (Democratic seats)
- ESS correction (G.4): $n^* = 695$ (not 10,000)

Table 1: S-track paper summary. Method and primary contribution for each paper.

Paper	Title	Method	Primary contribution
S.1	Hypothesis testing for gerrymandering	Permutation test + calibrated ensemble	Calibrated p-values from G.1 ensemble data
S.2	Bayesian redistricting	Bayesian inference Beta posterior	Bayesian Detection Score (BDS) for courts
S.3	Power analysis for ensemble sizing	Power-ESS formula (Section 3.2)	Min. ensemble sizes; 50-state table
S.4	Multiple testing FDR control	Benjamini-Hochberg FDR at $q = 0.10$	BH applied to L-track 1,050-test battery

- Calibrated p-value: $p_{\text{cal}} = 0.008$
- 95% CI on true percentile: $[0.2\%, 1.5\%]$

The calibrated p-value of 0.008 supports rejection at $\alpha = 0.05$ even after correction, but the confidence interval shows the claim is “at most 1.5th percentile” rather than exactly “0.4th percentile.”

4.2. S.2: Bayesian Redistricting

S.2 provides the Bayesian Detection Score (BDS) developed in the S.2 paper [Della-Libera, 2026b]. BDS converts the ensemble percentile and ESS into a posterior probability that the plan was intentionally selected for partisan advantage. For NC 2022 ($\hat{p} = 0.004$, $n^* \approx 695$): $\text{BDS} \approx 0.97$.

4.3. S.3: Power Analysis

S.3 derives minimum ensemble sizes for 80% power at $\alpha = 0.05$ to detect a 10pp partisan advantage. Key results (from the formula in Section 3):

- NC ($k = 14$, $\rho_1 = 0.87$): minimum $\approx 5,000$ steps
- CA ($k = 52$, $\rho_1 = 0.94$): minimum $\approx 30,000$ steps

Current practice (10,000 steps) gives 80% power for NC but only 45% for California.

4.4. S.4: Multiple Testing

S.4 applies the Benjamini-Hochberg procedure to the L-track’s 1,050-test battery (7 metrics $\times$ 50 states $\times$ 3 cycles). At $q = 0.10$, BH correction reduces significant findings from 147 (uncorrected) to 89. All clear gerrymanders (NC, WI, TX enacted) remain significant. The 58 demoted cases were at the margin of significance under the uncorrected test, suggesting they are likely false positives.

4.5. Integration

The three components integrate as follows in a complete analysis:

1. Generate ensemble with minimum size from S.3’s 50-state table
2. Certify convergence using G.4’s $\hat{R}$ and ESS diagnostics
3. Compute calibrated p-values using S.1’s ESS-corrected formula
4. Apply S.4’s BH procedure to the full test battery
5. Report BDS from S.2 as a court-facing summary

This pipeline converts the current practice (“plan at Xth percentile”) into litigation-grade inference (“plan is statistically significant at FDR $q = 0.10$, with 80% power to detect the observed effect size, $BDS = 0.97$ ”).

5. Conclusion

Redistricting analysis currently operates in an inferential gap: it produces descriptive percentile claims that lack the statistical rigour required for litigation-grade evidence. The four challenges identified in this paper—combinatorial test statistics, simulation-dependent null distributions, multiple testing, and small-tail ESS—are solvable with existing statistical methodology.

The three-component framework (calibrated null distribution, FDR correction, power analysis) bridges this gap. It is not a new statistical theory; it is an application of well-established methods (Benjamini-Hochberg, ESS correction, permutation tests) to the redistricting domain.

The S-track papers (S.1–S.4) implement each component with worked examples and provide reference implementations in BISECT. Together, they convert the redistricting ensemble from a descriptive tool into a rigorous inferential instrument—one that can support expert testimony meeting the Daubert standard and provide courts with the “judicially manageable standard” that *Rucho* found lacking in partisan gerrymandering claims.

The practical recommendation for redistricting practitioners: (1) certify ensemble convergence using G.4’s diagnostics before reporting any percentile claim; (2) use ESS-corrected sample sizes for confidence intervals; (3) apply BH correction when reporting multiple metrics; and (4) report power as a companion to any null-hypothesis test. These steps convert anecdotal evidence of extremity into statistically defensible claims.

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